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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 18	
TITLE : Java GUI Button Event Handling	

Problem Statement

Write a Java GUI program to demonstrate button click event handling.

Learning Objectives

To implement button click events using Java event handling mechanisms.

Questions:

1. Create a GUI calculator where buttons perform addition and subtraction.
2. Create a Bank Balance Calculator GUI application using Java Swing where the user enters the initial balance and transaction amount. Use buttons to perform addition (deposit) and subtraction (withdrawal) and display the updated balance.

Programme

1:

assignment18 > GUICalculator.java > GUICalculator > GUICalculator()

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class GUICalculator extends JFrame implements ActionListener {

    JTextField num1, num2;
    JButton add, subtract;
    JLabel result;

    GUICalculator() {

        setTitle(title: "GUI Calculator");
        setSize(width: 400, height: 250);
        setLayout(new GridLayout(rows: 4, cols: 2, hgap: 10, vgap: 10));

        num1 = new JTextField();
        num2 = new JTextField();
```

```java
        add = new JButton(text: "Addition");
        subtract = new JButton(text: "Subtraction");

        result = new JLabel(text: "Result: ");

        add(new JLabel(text: "First Number:"));
        add(num1);

        add(new JLabel(text: "Second Number:"));
        add(num2);

        add(add);
        add(subtract);

        add(new JLabel(text: ""));
        add(result);

        add.addActionListener(this);
        subtract.addActionListener(this);

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(b: true);
    }

    public void actionPerformed(ActionEvent e) {

        double a = Double.parseDouble(num1.getText());
        double b = Double.parseDouble(num2.getText());

        if (e.getSource() == add) {
            result.setText("Result: " + (a + b));
        }
        else if (e.getSource() == subtract) {
            result.setText("Result: " + (a - b));
        }
    }

    Run | Debug
    public static void main(String[] args) {

        new GUICalculator();
    }
}
```

2:

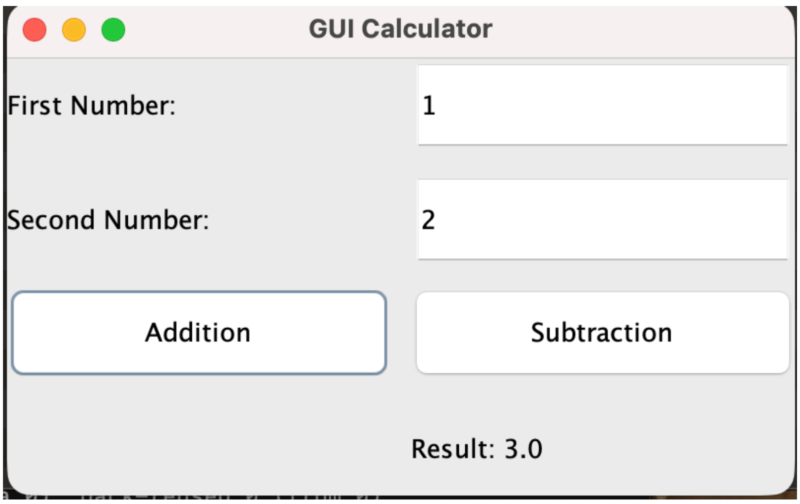
assignment18 > BankBalanceCalculator.java > BankBalanceCalculator > BankBalanceCalcula

```java
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
public class BankBalanceCalculator extends JFrame implements ActionListener {
    JTextField balanceField, transactionField;
    JButton deposit, withdraw;
    JLabel result;
    double balance;
    BankBalanceCalculator() {
        setTitle(title: "Bank Balance Calculator");
        setSize(width: 450, height: 300);
        setLayout(new GridLayout(rows: 4, cols: 2, hgap: 10, vgap: 10));
        balanceField = new JTextField();
        transactionField = new JTextField();

        deposit = new JButton(text: "Deposit");
        withdraw = new JButton(text: "Withdraw");

        result = new JLabel(text: "Updated Balance: ");

        add(new JLabel(text: "Initial Balance:"));
        add(balanceField);

        add(new JLabel(text: "Transaction Amount:"));
        add(transactionField);

        add(deposit);
        add(withdraw);

        add(new JLabel(text: ""));
        add(result);

        deposit.addActionListener(this);
        withdraw.addActionListener(this);

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(b: true);
    }

```

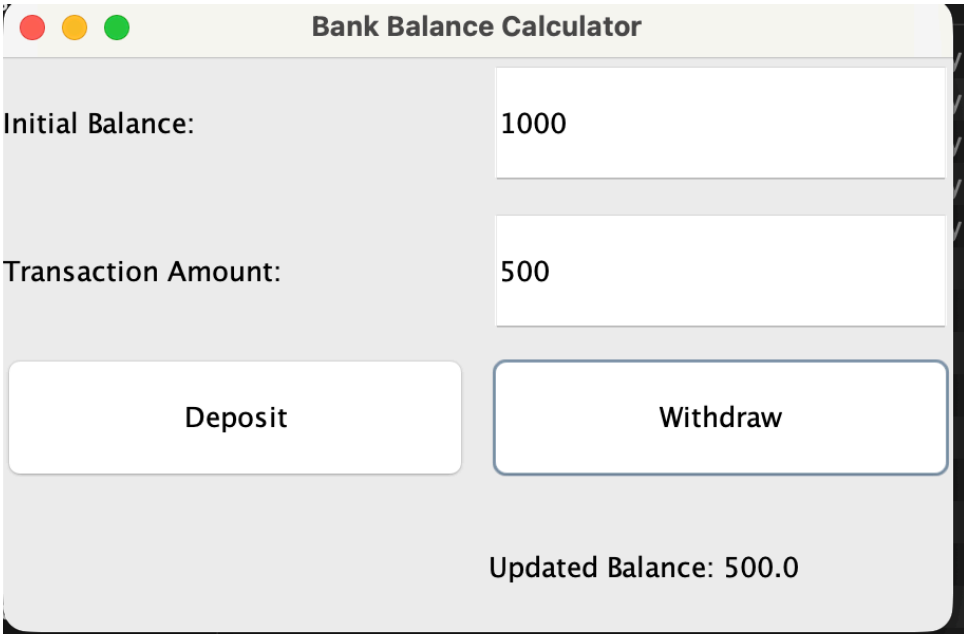
```java
    public void actionPerformed(ActionEvent e) {

        balance = Double.parseDouble(balanceField.getText());

        double amount =
            Double.parseDouble(transactionField.getText());

        if (e.getSource() == deposit) {

            balance = balance + amount;

        }
        else if (e.getSource() == withdraw) {

            balance = balance - amount;
        }

        result.setText("Updated Balance: " + balance);
    }

    Run | Debug
    public static void main(String[] args) {

        new BankBalanceCalculator();
    }
}
```

Output

1:



2:



Github:

Conclusion:

The programs successfully demonstrate button click event handling using Java Swing. JButton components generate action events , which are handled using ActionListener and actionPerformed(). The calculator performs addition and subtraction , while the bank application performs deposits and withdrawals and displays the updated balance.